Riccardo Pala

Software Engineer | Applied AI & RAG Systems

Software Engineer with 3+ years delivering reliable production systems and leading technical initiatives. M.Sc. in AI with hands-on experience building backend and retrieval infrastructure for RAG applications, including vector search and PostgreSQL/pgvector. Bridging software engineering, applied AI and technical coordination.

riccardo.pala98@outlook.it
+39 349 058 5195
Milan, Italy (Relocating to Barcelona)
linkedin.com/in/riccardo-pala
github.com/riccardo-pala

Professional Experience

Backend & RAG Infrastructure — Widiba PFM
Jun 2026 - Present

Silicon Shoring FS Reply • Banca Widiba • Milan

- **RAG Infrastructure:** Build backend services and data components supporting an LLM-powered RAG system for financial applications.
- **Vector Retrieval:** Implement hybrid structured/semantic search with PostgreSQL/pgvector, embedding-based retrieval, similarity ranking and relevance thresholds.
- **Technical Coordination:** Act as technical reference for the engineer developing the Python/LLM component, discussing implementation challenges and integration needs.
- **Domain Services:** Develop transaction-processing services for categorization, merchant normalization, geolocation and recurring-payment detection.

Senior Software Engineer & Technical Lead
Jun 2023 - Present

Silicon Shoring FS Reply • BPER Banca • Milan

- **Production Engineering:** Design and evolve Java/Spring REST services and distributed integrations for high-reliability financial platforms.
- **Data & Systems:** Develop Kafka event-driven services, improve database performance and troubleshoot production issues across cloud-native applications.
- **Technical Ownership:** Drive design discussions, reviews and engineering coordination while ensuring correctness, security and maintainability.

Education

M.Sc. Computer Science & Engineering (AI)
2020 - 2023

Politecnico di Milano

Final Grade: **110/110 cum laude**.

Specialization: Deep Learning, Machine Learning, Recommender Systems, Game Theory.

Research Exchange (XAI)
2022 - 2023

Universidad Politécnica de Madrid

Thesis: *Robustness of Ensemble Explanations (AXOM)* (Executive Summary).

Rated **10/10 (Top 5% of best thesis work)**.

Projects & Publications

ACM RecSys 2022 Conference Publication
Co-authored *"Lightweight Model for Session-Based Recommender Systems with Seasonality Information in the Fashion Domain"*. Explored neural-network and matrix-factorization approaches for session-based recommendation in Python.

eCommerce Online Learning System
Designed 6 online learning algorithms (UCB, Thompson Sampling) for behavior-based intelligent pricing.

Product Engineering (Clup & MuSeek)
Built end-to-end web and native Android products with J2E/Tomcat and Kotlin, including service-layer algorithms and API-backed application flows.

TECH STACK

Applied AI & Machine Learning

RAG Systems · Vector Retrieval · Embeddings · Semantic Search · Python · PyTorch · Deep Learning · Scikit-learn · Recommender Systems · XAI

Software Engineering

Java · Spring Boot · REST APIs · Microservices · System Design · SQL / pgvector

Cloud & Platform

OpenShift / Kubernetes · Docker · AWS · Apache Kafka · CI/CD · Dynatrace

CERTIFICATIONS

Confluent Certified Developer (CCDAK)
Confluent Certified Developer for Apache Kafka (Issued: Feb 2026)

AWS Certified AI Practitioner
IN PROGRESS
Expected: Q4 2026

LANGUAGES

English	Fluent (C1, TOEIC 955)
Spanish	Conversational (improving)
Italian	Native